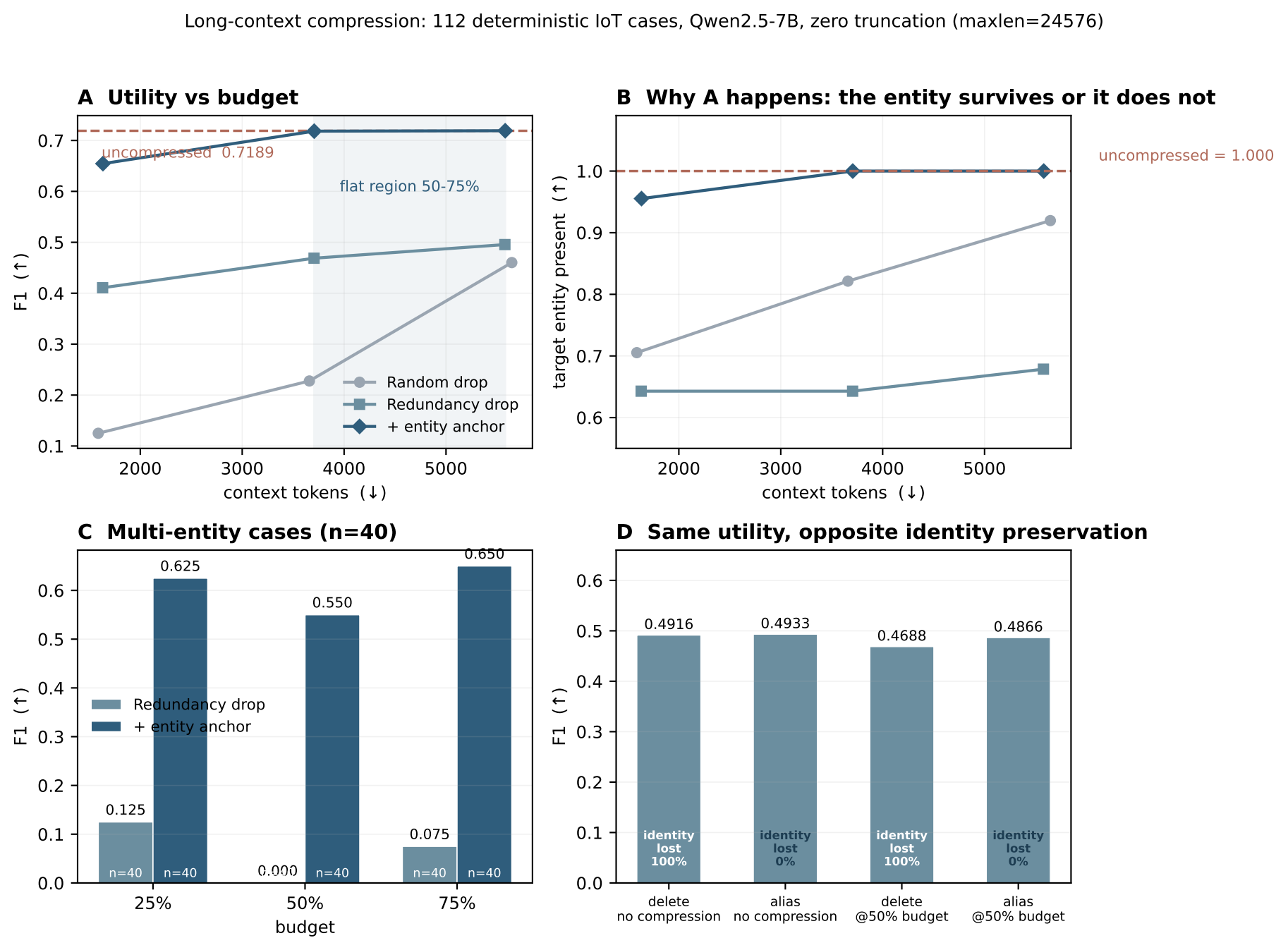
Long-context compression: 112 deterministic IoT cases, Qwen2.5-7B, zero truncation (maxlen=24576)
A Utility vs budget
uncompressed 0.7189
flat region 50-75%
Random drop
Redundancy drop
+ entity anchor
F1 (↑)
context tokens (↓)
B Why A happens: the entity survives or it does not
uncompressed = 1.000
target entity present (↑)
context tokens (↓)
C Multi-entity cases (n=40)
Redundancy drop
+ entity anchor
0.125
0.625
0.000
0.550
0.075
0.650
n=40
F1 (↑)
25%
50%
75%
budget
D Same utility, opposite identity preservation
0.4916
0.4933
0.4688
0.4866
identity lost 100%
identity lost 0%
identity lost 100%
identity lost 0%
F1 (↑)
delete no compression
alias no compression
delete @50% budget
alias @50% budget